SEMINAR WS 2026-2027

INTRODUCTION TO ALGEBRAIC K-THEORY

Organizers: Denis-Charles Cisinski, Giovanni Rossanigo

Time and place: Thursday, 14:15–15:45, M009

Overview

The goal of this seminar is to introduce basic concepts of algebraic K-theory. Algebraic K-theory is a general theory of finite measures generalizing the notions of rank of projective modules of finite type and of their determinant. It is at the core of the definition of many fundamental numerical invariants in algebraic geometry, with close relationship with Chow groups and thus with intersection theory, through the theory of Chern classes. The goal of the seminar is to establish on firm ground the basic definitions and prove their universal properties, starting from scratch. The first session of the seminar will be the opportunity to provide a detailed but broad overview. The exact contents of the seminar will be then be adapted to the interests of the audience.

Organization

Students taking the seminar for a grade will be assessed on their written report. They must send a preliminary draft of their report to the organizers at least ten days before their scheduled talk, allowing time for feedback and revisions. A final version must be submitted after the talk, incorporating feedback on the draft and any corrections or clarifications arising during the presentation and discussion. The report should give a self-contained account of the assigned material, with precise statements, detailed proofs of the main results, and references for results used without proof.

Talks

Talk 1: Introduction and distribution of talks (Denis-Charles Cisinski, 15.10.2026). Overview of the seminar and distribution of the talks. Please attend if you are considering giving a talk, otherwise contact the organizers.

Talk 2: Higher category theory I (Speaker, 22.10.2026). The goal of this talk is to introduce the categorical language used throughout the seminar. Follow [CCW26, Sections 1.1, 1.3, 1.7 and 2.1] to discuss categories, functors, functor categories, natural transformations and equivalences. Introduce composition through the Segal axiom and explain the meaning of the Rezk axiom [CCW26, Sections 2.2 and 2.3]. There is no need to develop the formal language of contexts or prove the foundational coherence results.

Introduce animae and the groupoid core of a category [CCW26, Definition 1.4.12 and Section 2.4]. Prove the universal property of the core [CCW26, Corollary 1.4.13] and explain the distinction between a category and its core, using the category $[1]$ as an example. Discuss the objectwise criterion for natural isomorphisms [CCW26, Proposition 3.4.6]; the foundational results used in its proof may be taken for granted. Conclude by introducing fully faithful functors [CCW26, Definition 5.1.1] and stating the fundamental theorem of category theory [CCW26, Theorem 5.3.3]. Explain the notion of strong surjectivity appearing in this statement. The proof may be omitted.

Talk 3: Higher category theory II (Speaker, 29.10.2026). The goal of this talk is to introduce universal constructions and the tools needed to define the $S_\bullet$-construction. Begin with adjunctions [CCW26, Section 4.4], explaining units, counits and the characterization through hom spaces [CCW26, Proposition 4.4.7]. Define limits and colimits following [CCW26, Section 4.11]. Discuss initial and

terminal objects, products and coproducts, and pullback and pushout squares. Prove that left adjoints preserve colimits and right adjoints preserve limits [CCW26, Lemma 4.11.6].

Introduce localizations through their universal property [CCW26, Definition 3.3.6], taking their existence for granted. Explain geometric realization as localization at all morphisms [CCW26, Definition 3.4.1], and distinguish it from the groupoid core discussed in the previous talk. Briefly explain reflective localizations and their relation to adjunctions [CCW26, Sections 4.5 and 13.2].

Introduce presheaves and state the Yoneda lemma [CCW26, Theorem 11.3.16]. Explain its evaluation map and deduce the full faithfulness of the Yoneda embedding; the synthetic proof of the theorem may be omitted. Finally, introduce left and right Kan extensions as adjoints to restriction and state the pointwise formulas [CCW26, Theorems 12.1.8 and 12.1.9]. Illustrate left Kan extension by completing a span to a pushout square. Define simplicial objects and their geometric realizations, and explain the distinction between these realizations and the realization of a category.

Talk 4: Waldhausen categories (Speaker, 05.11.2026). The goal of this talk is to introduce the categories on which we will construct algebraic K-theory. Begin with pointed categories and zero objects. Introduce semiadditive and additive categories following [CCW26, Definitions 19.2.3 and 19.2.7], and explain the addition of morphisms [CCW26, Construction 19.2.4]. Use finite-dimensional vector spaces as a familiar example.

Define pre-Waldhausen categories, Waldhausen categories and their morphisms [CCW26, Definition 19.3.1]. Explain the roles of cofibrations, weak equivalences and the gluing axiom. Give the examples of finite pointed sets with injections and isomorphisms. Also discuss the Waldhausen structure on a pointed category with finite colimits [CCW26, Example 19.3.4].

Prove [CCW26, Lemma 19.3.2] and use it to explain the Waldhausen structure on categories of exact functors [CCW26, Construction 19.3.3]. Work out explicitly the case of the category of cofibrations, including its cofibrations and weak equivalences. Conclude with the pointwise Waldhausen structure on functor categories and the categories $w_n(A)$, [CCW26, Example 19.3.5 and Notation 19.3.6], which will be used in the next talk.

Talk 5: Construction of K-theory (Speaker, 12.11.2026). The goal of this talk is to construct the K-theory space of a Waldhausen category. First introduce short exact sequences and the category $\mathrm{ExSeq}(A)$ [CCW26, Definition 19.4.1]. Prove that the category of Waldhausen categories is semiadditive [CCW26, Proposition 19.4.4], using [CCW26, Lemmas 19.4.2 and 19.4.3]. Introduce Waldhausen regions and the categories $S(R, A)$ following [CCW26, Definitions 19.5.1 and 19.5.4], concentrating on the arrow categories $\mathrm{Fun}([1], [n])$. Explain their Waldhausen structures [CCW26, Construction 19.5.6] and define the simplicial Waldhausen category $S_\bullet(A)$ [CCW26, Definition 19.5.7]. Prove [CCW26, Lemma 19.5.8], identifying these diagrams with filtrations by cofibrations. Describe explicitly $S_0(A)$, $S_1(A)$ and $S_2(A)$, and explain the face and degeneracy maps.

Finally, incorporate weak equivalences and define the K-theory space by the iterated realization and loop construction [CCW26, Remark 19.5.9 and Definition 19.5.10]. Explain its functoriality and the simplification when weak equivalences are precisely the isomorphisms.

Talk 6: Additivity of Waldhausen K-theory I (Speaker, 19.11.2026).

Talk 7: Additivity of Waldhausen K-theory II (Speaker, 03.12.2026).

Talk 8: Stable categories (Speaker, 10.12.2026). Introduce stable categories following [CCW26, Definition 18.1.1]. Explain the relation between suspension, loop objects, fibres and cofibres, and state that a stable category admits finite limits and finite colimits and that a square is cartesian if and only if it is cocartesian [CCW26, Theorem 18.3.3]. Prove the equivalent characterizations of exact functors between stable categories [CCW26, Theorem 18.3.4]. Discuss the examples of spectra and derived categories of modules, and explain how a stable category becomes a Waldhausen category by taking all morphisms as cofibrations and equivalences as weak equivalences.

Introduce retracts and idempotent completeness [CCW26, Definitions 16.4.1 and 16.4.6]. State the existence and universal property of the idempotent completion of a small stable category, including its description as the compact objects of its Ind-completion [BGT13, Lemma 2.20]. Explain why a full stable subcategory of an idempotent-complete stable category is idempotent complete precisely when it is closed under retracts.

For a fully faithful exact functor between small stable categories $\mathcal{A} \to \mathcal{B}$, introduce the Verdier quotient $\mathcal{B}/\mathcal{A}$ through its universal property [CCW26, Definition 18.6.2]. Prove that it is obtained by inverting the morphisms whose cofibres belong to $\mathcal{A}$ [CCW26, Proposition 18.6.7], taking the existence and stability of the required localization from [CCW26, Proposition 18.6.1]. Introduce thick subcategories and state that $\mathcal{A}$ is the kernel of $\mathcal{B} \to \mathcal{B}/\mathcal{A}$ precisely when it is thick [CCW26, Definition 18.6.16 and Proposition 18.6.17]. Conclude by distinguishing the Verdier quotient $\mathcal{B}/\mathcal{A}$ from its idempotent completion $(\mathcal{B}/\mathcal{A})^{\natural}$, which will enter the non-connective localization theorem.

Talk 9: Localization and non-connective K-theory (Speaker, 17.12.2026).

Talk 10: Derived categories of schemes (Speaker, 07.01.2027). The goal of this talk is to define the K-theory and G-theory of schemes. Recall the derived category of modules over a ring and introduce quasi-coherent complexes on a scheme following [Kha18, Lecture 1, Section 3], restricting to ordinary schemes. State Zariski descent and deduce the gluing description for a cover by two open subschemes [Kha18, Lecture 2, Corollary 1.7 and Theorem 3.2]. There is no need to introduce derived schemes or construct categories of module spectra.

Introduce perfect complexes, explaining their description on affine schemes as bounded complexes of finitely generated projective modules [Kha18, Lecture 1, Remark 3.8]. Prove their characterization as compact objects in the affine case [Kha18, Lecture 1, Exercise 3.9], and state the corresponding result for quasi-compact and quasi-separated schemes [Kha18, Lecture 3, Propositions 3.5 and 3.10]. Define the K-theory of a scheme using perfect complexes. For a noetherian scheme, introduce bounded complexes with coherent cohomology and define G-theory. State the comparisons with the classical definitions [TT90, Proposition 3.10 and Corollary 3.13], and explain how to recover the usual Grothendieck groups. Conclude with functoriality under pullback, distinguishing the hypotheses needed for K-theory and G-theory [TT90, Section 3.14].

Talk 11: Localization in algebraic geometry (Speaker, 14.01.2027). The goal of this talk is to prove the localization sequence associated with an open immersion. Let X be a quasi-compact and quasi-separated scheme and let $U \subseteq X$ be a quasi-compact open subscheme. Introduce quasi-coherent complexes and perfect complexes supported on the complement, and explain the localization sequence for quasi-coherent complexes using restriction and derived direct image.

Prove compact generation with support in the affine case [Kha18, Lecture 3, Proposition 5.2]. Construct the relevant Koszul complex and show that its shifts detect zero objects. State the global version [Kha18, Lecture 3, Theorem 6.2]; its proof may be omitted. Using the localization theorem for compact objects [Kha18, Lecture 3, Proposition 2.4], identify the category of perfect complexes on U with the idempotent completion of the appropriate Verdier quotient. Explain why idempotent completion is necessary.

Apply the localization theorem from Talk 9 to obtain the geometric localization sequence [TT90, Theorem 7.4]. Work out the example of a principal open subscheme of an affine scheme. Conclude by proving Zariski excision for perfect complexes with support, using the gluing result from Talk 10. This will be used to establish descent in Talk 13.

Talk 13: Nisnevich descent (Speaker, 28.01.2027). The goal of this talk is to deduce Nisnevich descent from localization and excision. Begin by proving the Mayer-Vietoris sequence for a cover by two quasi-compact open subschemes [Kha18, Lecture 4, Theorem 3.2]. Use geometric localization and the equivalence of categories with support established in Talk 11. Explain why descent for perfect complexes alone does not formally imply descent for their K-theory.

Introduce the Nisnevich topology and distinguished Nisnevich squares [Kha18, Lecture 5, Section 7]. Give examples and explain the condition that the étale morphism induces an isomorphism over the complement of the open subscheme. Establish excision for perfect complexes with support in this situation, following the étale case of [TT90, Theorem 2.6.3]. Give the affine argument and explain the reduction from schemes to this case. Flat base change and étale descent for quasi-coherent complexes may be used without proof.

Compare the localization sequences on the two sides of a distinguished square to prove [Kha18, Lecture 5, Theorem 7.3]. State the criterion relating distinguished squares to Nisnevich descent [Kha18, Lecture 5, Remark 7.4], without proving the general criterion. Conclude that non-connective K-theory satisfies Nisnevich descent.

Talk 14: Projective bundle formula (Speaker, 05.02.2027). The goal of this talk is to prove the projective bundle formula using additivity. Work with a vector bundle of finite positive rank on a quasi-compact and quasi-separated scheme. Recall the required cohomology computations for the twisting sheaves on its projective bundle [Kha20, Proposition 3.2]. Explain semiorthogonal decompositions and deduce their effect on K-theory from additivity [Kha20, Lemmas 2.5 and 2.8].

Prove the decomposition of the category of perfect complexes on the projective bundle [Kha20, Theorem B], following the proof of [Kha20, Theorem 3.3 and Section 3.3]. Check full faithfulness and semiorthogonality using the cohomology computations. For generation, explain the Koszul relation [Kha20, Lemma 3.4] and the successive evaluation maps used in the proof. The generation statement [Kha20, Lemma 3.5] and preservation of perfect complexes by the derived direct image may be assumed.

Apply additivity to prove the projective bundle formula [Kha20, Corollary 3.6], and compare its formulation with [TT90, Theorems 4.1 and 7.3]. Describe explicitly the summands given by pullback and tensoring with the twisting line bundles. Conclude by computing the K-groups of projective space over a field. If time permits, determine its Grothendieck ring using the Koszul relation.

References

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